Recording L01-R01
Quadratic separation of a Gaussian

A proof of the formula for the conditional density of a Gaussian

Probabilistic Machine Learning 2027-1

2026-10-17

From the lecture to this proof

From the lecture L01: Gaussian blocks, conditioning and centered residual

$$X = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \sim \mathcal{N}_d(\mu, \Sigma)$$

$$\mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}$$

$$\Sigma = \Sigma^T \succ 0, \quad d = d_1 + d_2$$

To prove here

$$p(x_1|x_2) = \mathcal{N}_{d_1}(x_1; m(x_2), S)$$

The conditional mean $m(x_2)$ and covariance S obtained in the lecture. Derive this complete density from the supplied joint density $p(x_1, x_2)$.

Fix the observed value x_2; the density describes the missing block X_1.
We account for the exponential and its coefficient, then identify the conditional by integration and division.

The density contains an exponential and a coefficient

$$p(x_1, x_2) = C \exp\left(-\frac{1}{2} Q\right)$$

The quadratic in the exponent

$$Q = (x - \mu)^T \Sigma^{-1} (x - \mu)$$

Target: a residual contribution plus an observed-deviation contribution.

Residual: $x_1 - m(x_2)$. Observed deviation: $x_2 - \mu_2$.

> Writing the exponential as a product will not establish a full density product until the coefficient also matches.

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The coefficient before the exponential

$$C = (2\pi)^{-\frac{d}{2}} (\det \Sigma)^{-\frac{1}{2}}$$

$$d = d_1 + d_2$$

Target: the coefficient for the residual density times that for the observed-block density.

> Writing the exponential as a product will not establish a full density product until the coefficient also matches.

Choose B to remove the cross covariance

Subtract B times the observed block so the cross covariance cancels:

$$H := \Sigma_{22}$$

$$\Sigma_{12} - BH = 0 \quad \Leftrightarrow \quad BH = \Sigma_{12}$$

Canceling the cross blocks makes the inverse act separately on residual and observed coordinates. This is what will give the two terms in Q.

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Why the inverse of H exists

Take any nonzero vector z for the observed block:

$$z \in \mathbb{R}^{d_2} \setminus \{0\}$$

$$z^T H z = \begin{pmatrix} 0 \\ z \end{pmatrix}^T \Sigma \begin{pmatrix} 0 \\ z \end{pmatrix} > 0$$

$$H = H^T \succ 0 \quad \Rightarrow \quad H^{-1} \text{ exists}$$

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Solve the cancellation equation

$$B = \Sigma_{12} H^{-1} \in \mathbb{R}^{d_1 \times d_2}$$

$$BH = \Sigma_{12}, \quad HB^T = \Sigma_{21}$$

Canceling the cross blocks makes the inverse act separately on residual and observed coordinates. This is what will give the two terms in Q.

Subtracting the predicted deviation produces the residual

$$u := x_2 - \mu_2$$

$$m(x_2) := \mu_1 + Bu, \quad r := x_1 - m(x_2)$$

Here r is the error around the predicted mean and u is the observed deviation. The target uses the residual covariance S from the lecture; it will be derived here.

Subtracting the predicted deviation produces the residual

$$u := x_2 - \mu_2$$

$$m(x_2) := \mu_1 + Bu, \quad r := x_1 - m(x_2)$$

$$T := \begin{pmatrix} I_{d_1} & -B \\ 0 & I_{d_2} \end{pmatrix}$$

$$T(x - \mu) = \begin{pmatrix} I & -B \\ 0 & I \end{pmatrix} \begin{pmatrix} x_1 - \mu_1 \\ u \end{pmatrix} = \begin{pmatrix} r \\ u \end{pmatrix}$$

Target from the lecture — to prove here:

$$Q = r^T S^{-1} r + u^T H^{-1} u$$

Here r is the error around the predicted mean and u is the observed deviation. The target uses the residual covariance S from the lecture; it will be derived here.

The coordinate change is reversible

The map sent $(x_1 - \mu_1, u)$ to (r, u); add the predicted deviation back:

$$U := \begin{pmatrix} I & B \\ 0 & I \end{pmatrix}, \quad U \begin{pmatrix} r \\ u \end{pmatrix} = \begin{pmatrix} r + Bu \\ u \end{pmatrix} = x - \mu$$

The residual and observation together determine the original centered vector. Invertibility will preserve strict positivity and let us recover the inverse covariance.

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$$TU = \begin{pmatrix} I & B - B \\ 0 & I \end{pmatrix} = I$$

$$UT = \begin{pmatrix} I & -B + B \\ 0 & I \end{pmatrix} = I \quad \Rightarrow \quad T^{-1} = U$$

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Subtract the observed rows to cancel the upper-right block

Using $BH = \Sigma_{12}$, multiply by T on the left:

$$\Sigma = \begin{pmatrix} \Sigma_{11} & BH \\ HB^T & H \end{pmatrix}, \quad T = \begin{pmatrix} I & -B \\ 0 & I \end{pmatrix}$$

The chosen B cancels the upper-right block exactly. The lower-left block remains because only the row operation has been applied.

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$$S := \Sigma_{11} - BHB^T, \quad T\Sigma = \begin{pmatrix} S & 0 \\ HB^T & H \end{pmatrix}$$

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